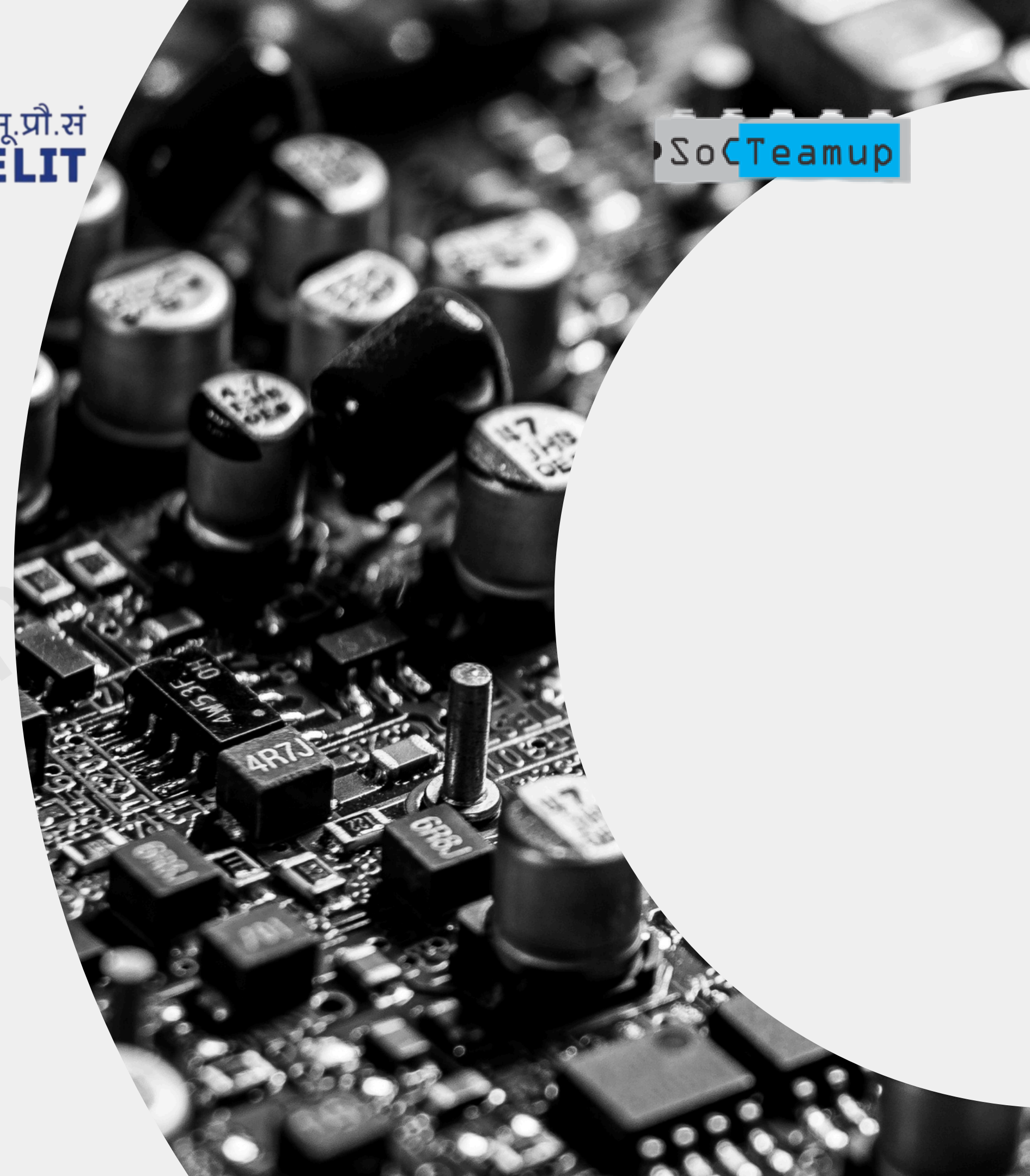


PVT (Process, Voltage, Temperature) Corner Analysis.





Outline

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2	Why PVT Corners are Important
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➔ What is PVT

PVT Stands for:

- Process, Voltage and Temperature.
- These are the three major factors that affect a chip's timing, power, and reliability.
- Chips don't always operate at ideal conditions.

➔ Why **PVT** Corners are Important

PVT corner analysis helps:

- Ensure functionality in all operating conditions
- Avoid timing and power failures
- Meet yield and reliability goals

➔ Process Variation (P)

Process variation occurs due to:

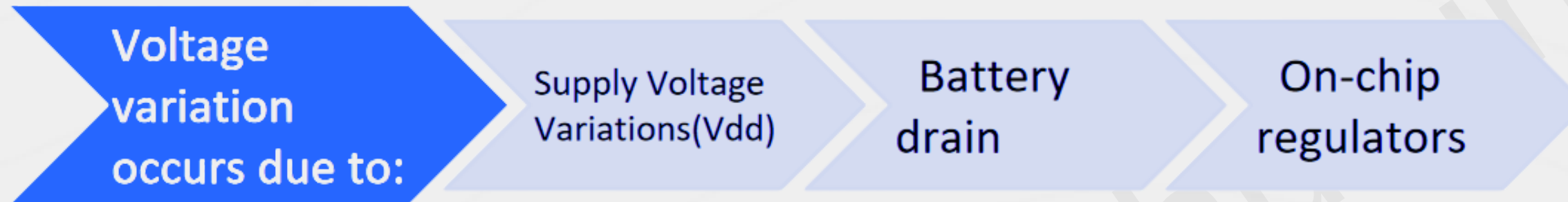
- Manufacturing variations in the semiconductor fabrication process.

Due to fabrication differences, transistors can be:

- Slow-Slow (SS): Slow NMOS and PMOS
- Fast-Fast (FF): Fast NMOS and PMOS
- Typical-Typical (TT): Nominal case

Impacts both delay and leakage power.

➔ Voltage Variation (V)



Common V ranges:

- 0.95V to 1.10V (for nominal 1.0V designs)

Lower voltage → slower switching
Higher voltage → faster but higher power

➔ Temperature Variation (T)



Range: -40°C to 125°C (typical for automotive)

- Higher T → slower speed, higher leakage
- Lower T → faster switching, lower leakage

What is a **PVT** Corner?

PVT Corner = A combination of Process, Voltage, and Temperature

Example:

FF@1.10V@0°C → Fastest case

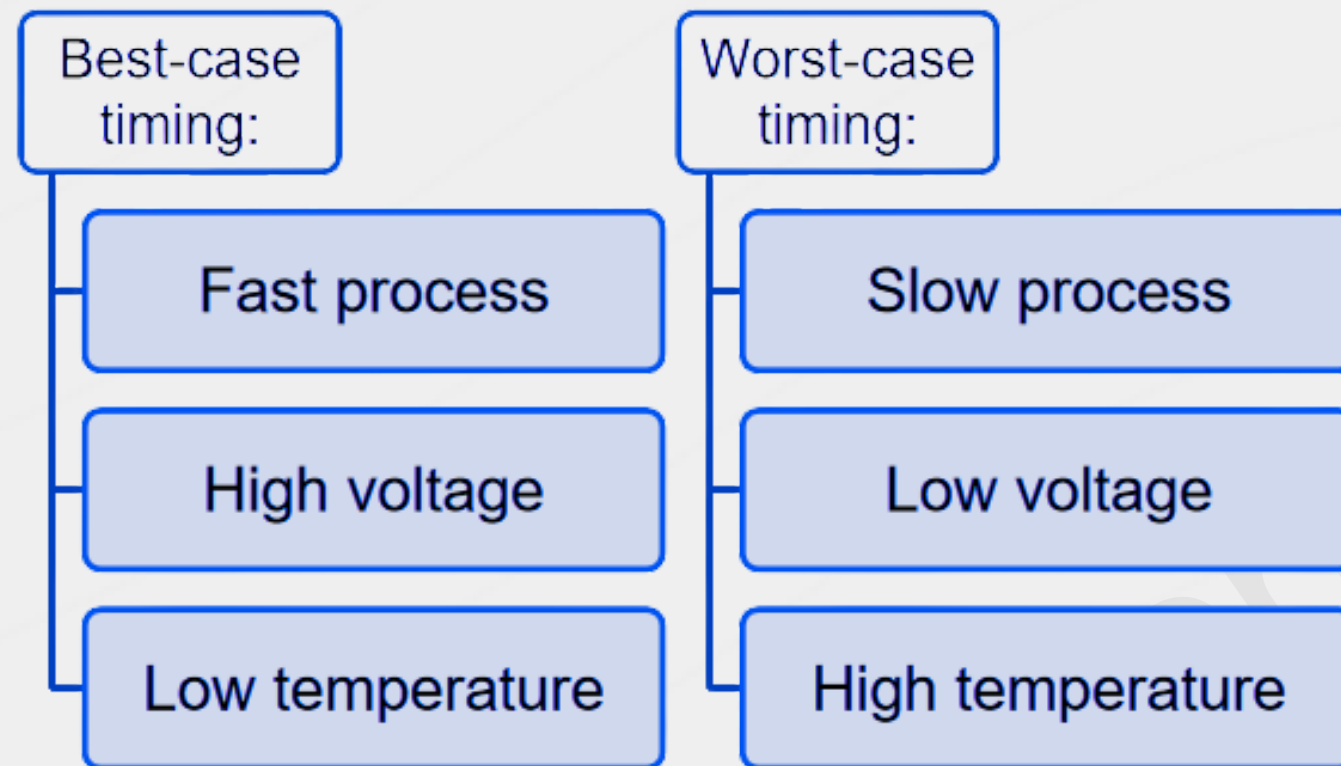
SS@0.95V@125°C → Slowest case

Example:

Slow Process + Low Voltage + High Temperature

Each corner models a worst-case or best-case operating condition.

➔ Best Case vs. Worst Case Timing



STA checks setup/hold at all timing corners.

Setup violations → worst-case delay
Hold violations → best-case delay

➔ How PVT Affects Delay and Power

P affects:

- Gate delay, leakage power

V affects:

- Delay (higher V = faster)
- Dynamic power ($\propto V^2$)

T affects:

- Delay, leakage, IR drop

PVT corners guide timing, power, and reliability analysis.

➔ STA at Multiple Corners

Static Timing Analysis (STA) must be run at:

- Multiple PVT combinations
- Setup/hold checks under all cases

Modern flows:

Multi-Corner Multi-Mode (MCMM) analysis

Simultaneous analysis of multiple PVT conditions

Results can vary widely per corner, so worst-case slack matters most.

Library Files per Corner

Each corner has its own `.lib` timing file:

- ss_0p95v_125c.lib
- ff_1p05v_0c.lib

The synthesis and STA tools use the correct `.lib` for each corner.

Corner-specific `.spef`, `.sdf`, and parasitics may also be used

➔ **Timing Closure Under PVT Corners**

Design must close timing under:

- All relevant corners (typical, slow, fast)
- Both setup and hold paths

Designers buffer, size, and pipeline to meet this goal.

Failing one corner may result in silicon failure.

Challenges in PVT Corner Analysis

Challenges

- Large number of timing path
- Increased tool runtime
- Corner-specific fixes needed
- Conflicting tradeoffs (e.g., speed vs. leakage)

**Automated corner closure techniques
are evolving.**

Chip signoff Requirements

Before tape-out

- Setup clean at worst corner
- Hold clean at best corner
- IR drop verified
- EM/IR analysis done
- Power analysis complete
- Functional verification across corners

Signoff requires passing all required PVT corners.

➔ Summary and Best Practices

Summary:

- PVT = Process, Voltage, Temperature
- Corners model real-world manufacturing and operating conditions
- STA must check all PVT corners

Best Practices:

- Simulate at all worst-case corners
- Ensure proper `.lib` mapping
- Check both setup and hold paths per corner

Thank You

echiphu

